INCIDENT ANALYSIS REPORT

Task # 04: Advanced Dynamic Analysis and Sandbox Evasion of Agent Tesla

SUBMITTED BY:

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INCIDENT ANALYSIS REPORT **1**
Task # 04: Advanced Dynamic Analysis and Sandbox Evasion of Agent Tesla 1
EXECUTIVE OVERVIEW 2
TARGET ENVIRONMENT & CLOAKING ARCHITECTURE 3
FORENSIC STACK & TOOLS DEPLOYED 3
Step 1: The Lockdown (Preventing Escape) 4
Step 2: The Cloak (Tricking Agent Tesla) 7
1. Process Explorer & Process Monitor (ProcMon) 10
2. Regshot 11
3. FakeNet-NG 12
4. DumpIt 13
5. Netstat 14
How to Download Agent Tesla safely inside the VM 14
Phase 1: The Pre-Detonation Setup (Starting the Traps) 16
Phase 2: The Detonation 20
Phase 3: The Observation (Do not touch the mouse for 3 minutes!) 20
Phase 4: Data Collection (Securing the Evidence) 21
Are the Evidence Files Infected? 22
Step 1: Package the Evidence (Inside the VM) 23
Step 2: Open the Secure Door (On Kali Linux) 23
Step 3: Push the Evidence Out (Inside the VM) 23
Step 4: NUKE IT FROM ORBIT (The Revert) 24

EXECUTIVE OVERVIEW

This report details the advanced dynamic analysis and behavioral profiling of **Agent Tesla**, a sophisticated Information Stealer and Keylogger. Because Agent Tesla actively employs anti-analysis and sandbox-evasion techniques (specifically querying WMI for virtualization artifacts), the analysis environment required significant hardening. A fully cloaked, isolated Windows 10 virtual environment was engineered by modifying hypervisor configuration files and disabling guest integration features to trick the malware into full execution.

Following a successful detonation, a comprehensive suite of forensic tools was utilized to capture the malware's process injection, registry modifications, memory footprint, and simulated command-and-control (C2) network traffic. All forensic artifacts were securely exfiltrated to a secure Kali Linux host via an encrypted SSH tunnel before reverting the environment to a pristine state.

TARGET ENVIRONMENT & CLOAKING ARCHITECTURE

- **Target Endpoint:** Windows 10 Virtual Machine (Isolated)
- **Hypervisor:** VMware (Cloaked via `.vmx` modification to disable backdoor and version reporting)
- **Network Posture:** Host-Only (Air-gapped from live internet)
- **Analysis Host:** Kali Linux (Secure receiving host for forensic artifacts)

FORENSIC STACK & TOOLS DEPLOYED

- **Malware Sample:** Agent Tesla (Sourced securely via MalwareBazaar / abuse.ch)
- **Network Simulation & Interception:** FakeNet-NG (Mandiant/FireEye)
- **Process Lineage & API Monitoring:** Microsoft Sysinternals Process Explorer & Process Monitor (ProcMon)
- **Registry Snapshot & Differential Analysis:** Regshot
- **Physical Memory (RAM) Acquisition:** Magnet DumpIt
- **Network Connection Mapping:** Windows Native Netstat
- **Secure Artifact Exfiltration:** Native Windows SCP (Secure Copy Protocol) over SSH

DATE: March 13, 2026

Agent Tesla is an absolutely brilliant choice for this task. It is a classic, highly documented Information Stealer and Keylogger, which means you will have endless data to capture with Process Monitor and FakeNet-NG.

However, Agent Tesla is notoriously sneaky. Not only do we need to lock down VMware so the malware doesn't escape and nuke your Kali machine, but we also have to trick Agent Tesla into thinking it is running on a real computer. Advanced malware like Agent Tesla actually scans the

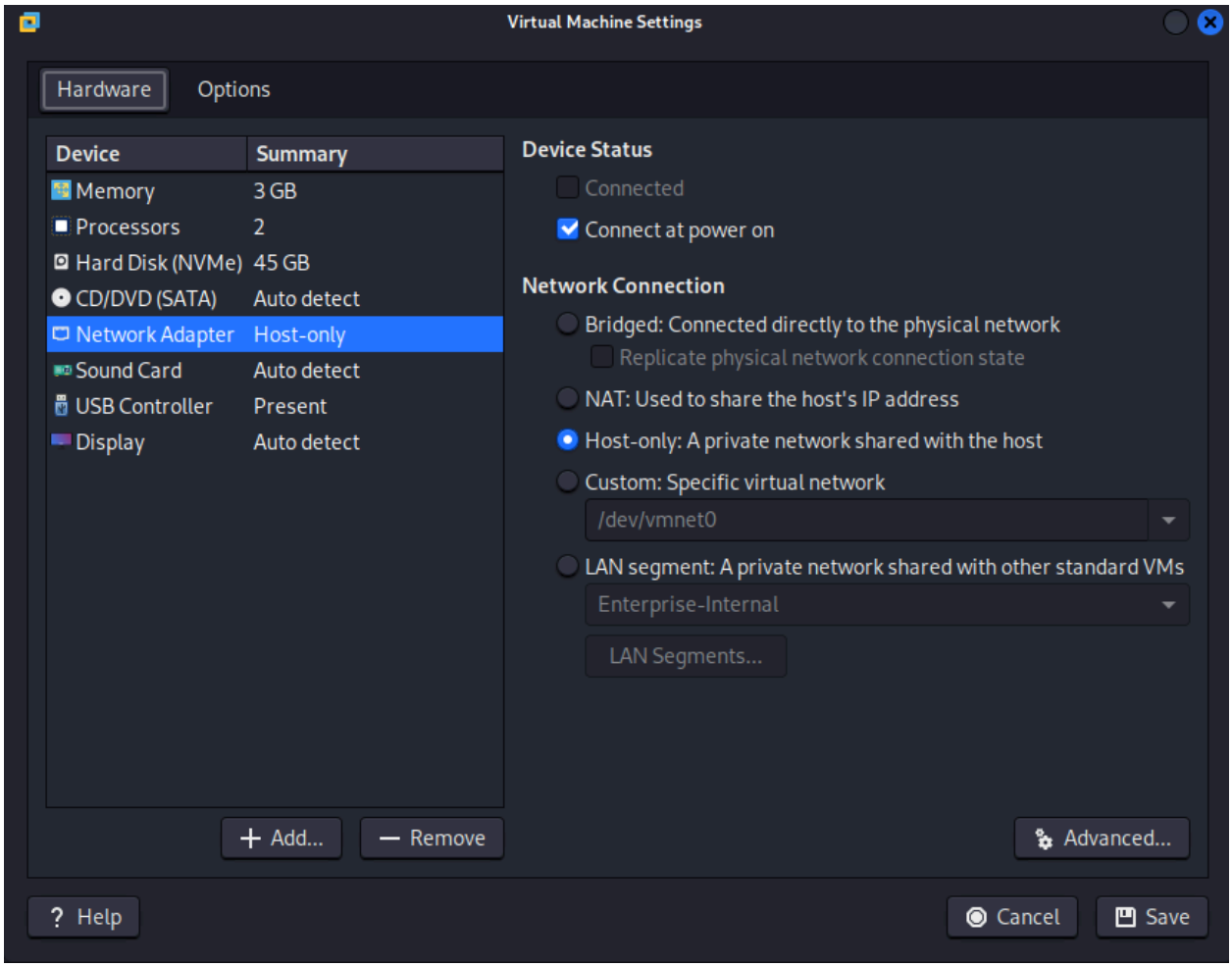
system to see if it is inside VMware. If it detects VMware, it will immediately kill itself and hide its behavior, ruining your analysis!

Here is exactly how to build an inescapable, invisible fortress in VMware:

Step 1: The Lockdown (Preventing Escape)

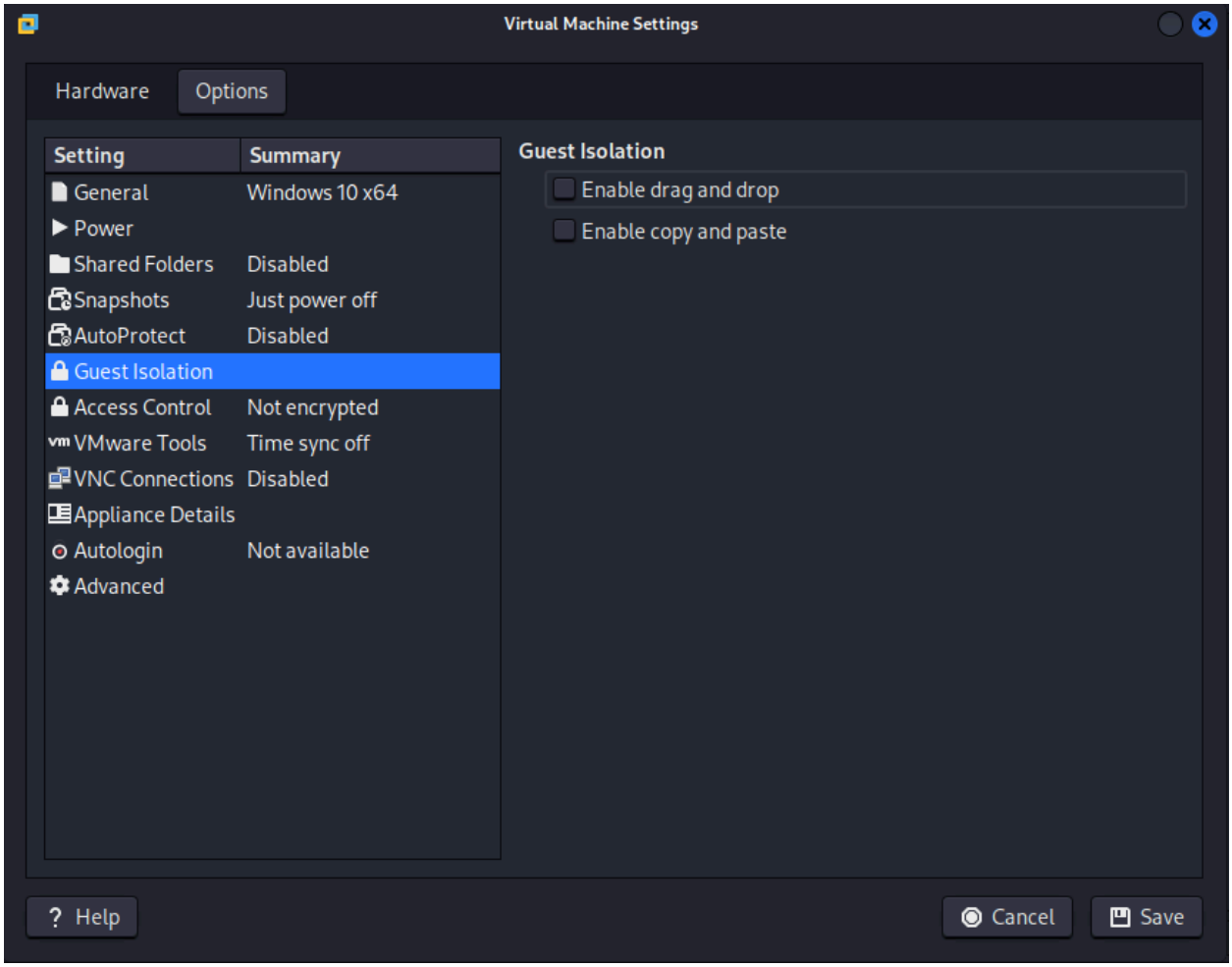
Before you even copy the malware to the VM, open your VMware settings for your Windows 10 VM and make these changes:

- **Network Adapter:** Change this from "Bridged" or "NAT" to "Host-Only" or a custom isolated VMnet. This ensures the malware cannot actually reach the real internet or your local Kali network. (Your tool, FakeNet-NG, will simulate the fake internet for it locally).

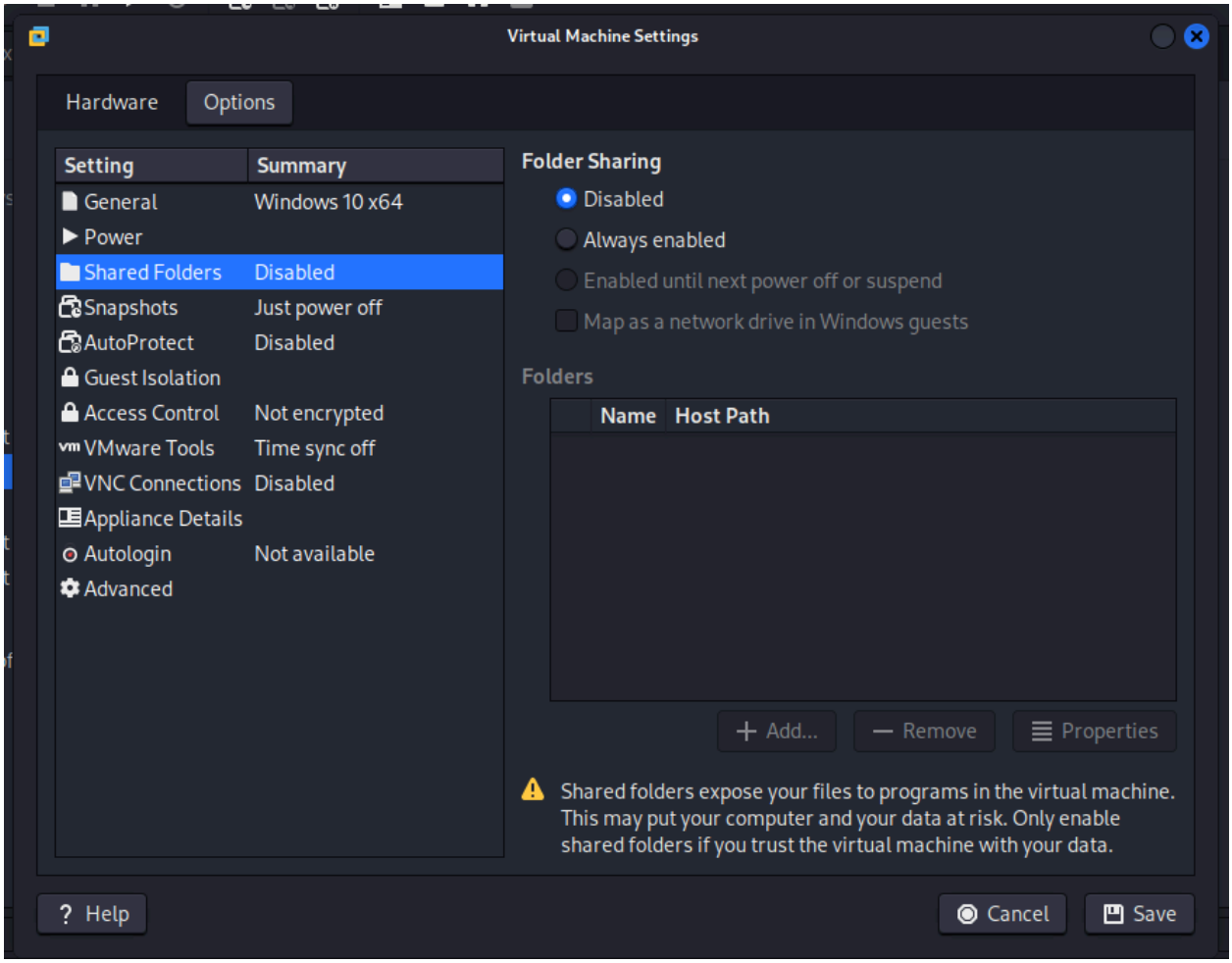


- **Guest Isolation:** Go to Options > Guest Isolation. Uncheck both Enable drag and drop and Enable copy and paste. (Malware can ride the shared clipboard right into your host

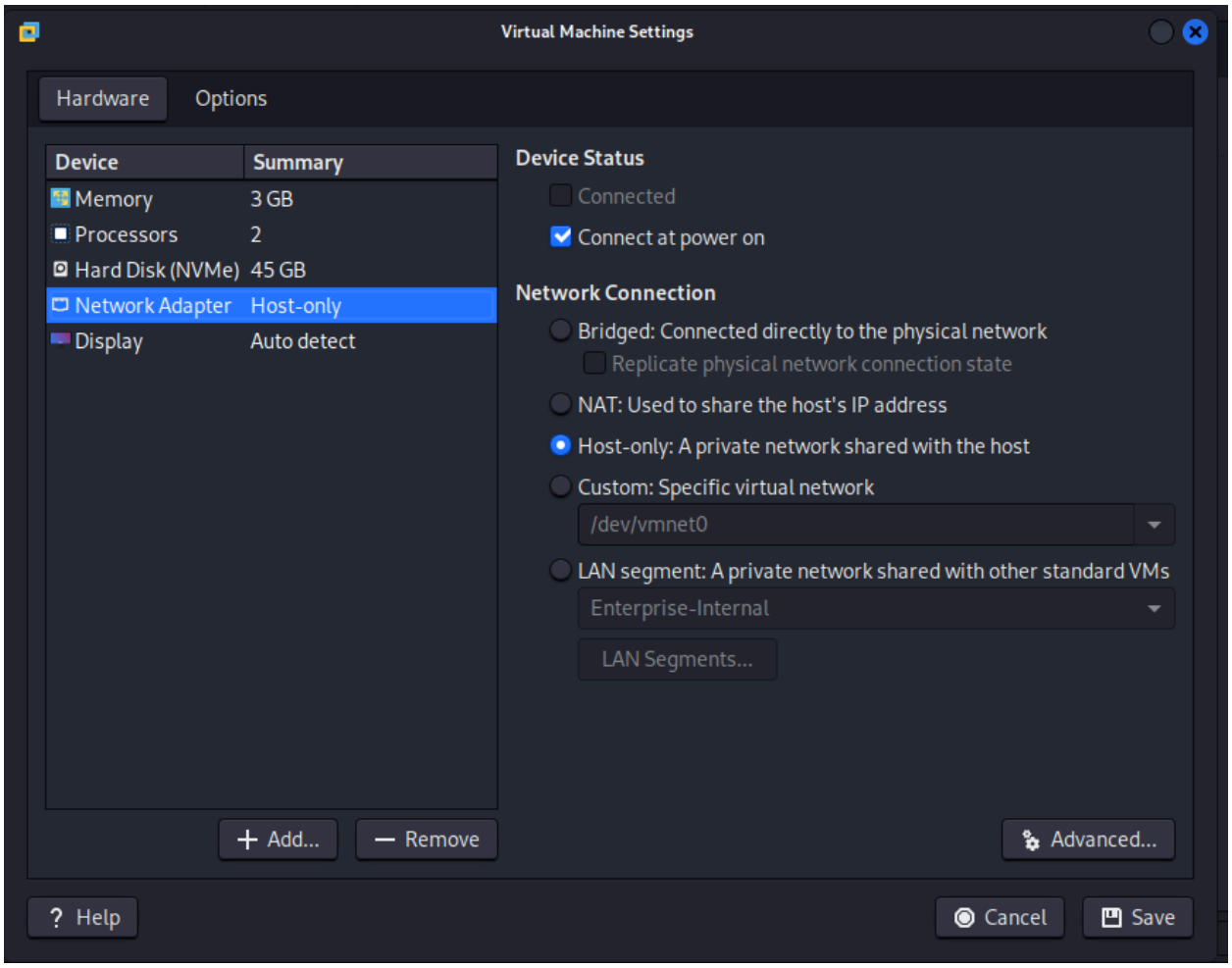
machine).



- **Shared Folders:** Go to Options > Shared Folders. Set this strictly to Disabled. Never share a Kali folder directly with a malware analysis VM.


Virtual Machine Settings
Hardware
Options
Setting
Summary
General
Windows 10 x64
Power
Shared Folders
Disabled
Snapshots
Just power off
AutoProtect
Disabled
Guest Isolation
Access Control
Not encrypted
VMware Tools
Time sync off
VNC Connections
Disabled
Appliance Details
Autologin
Not available
Advanced
Folder Sharing
Disabled
Always enabled
Enabled until next power off or suspend
Map as a network drive in Windows guests
Folders
Name
Host Path
Add...
Remove
Properties
Shared folders expose your files to programs in the virtual machine. This may put your computer and your data at risk. Only enable shared folders if you trust the virtual machine with your data.
Help
Cancel
Save

- **Hardware:** Remove the USB Controller, Printer, and Sound Card from the VM hardware list. The smaller the attack surface, the better.



Step 2: The Cloak (Tricking Agent Tesla)

Agent Tesla specifically queries Windows Management Instrumentation (WMI) to check if the hardware manufacturer is "VMware". We need to strip the VMware branding out of the machine's configuration.

1. Open your Linux terminal.
2. Navigate to where your VM is stored. By default, VMware usually saves these in your home directory under `vmware`. Type: `cd ~/vmware/` and press Enter.
3. Use the `ls` command to find the folder with your Windows 10 VM, and `cd` into it.

```
(base) ┌──(bhai㉿bhai)-[~/vmware/Windows 10 x64]
└─$ ls
 mksSandbox                          'Windows 10 x64-s004.vmdk'
 mksSandbox-0.log                    'Windows 10 x64-s005.vmdk'
 mksSandbox-1.log                    'Windows 10 x64-s006.vmdk'
 mksSandbox-2.log                    'Windows 10 x64-s007.vmdk'
 mksSandbox.log                      'Windows 10 x64-s008.vmdk'
 vmware-0.log                        'Windows 10 x64-s009.vmdk'
 vmware-1.log                        'Windows 10 x64-s010.vmdk'
 vmware-2.log                        'Windows 10 x64-s011.vmdk'
 vmware.log                          'Windows 10 x64-s012.vmdk'
'Windows 10 x64-0.scoreboard'        'Windows 10 x64.scoreboard'
'Windows 10 x64-1.scoreboard'        'Windows 10 x64.vmdk'
'Windows 10 x64-2.scoreboard'        'Windows 10 x64.vmdk.lck'
'Windows 10 x64.nvram'               'Windows 10 x64.vmsd'
'Windows 10 x64-s001.vmdk'           'Windows 10 x64.vmx'
'Windows 10 x64-s002.vmdk'           'Windows 10 x64.vmxf'
'Windows 10 x64-s003.vmdk'           'Windows 10 x64.vmx.lck'
```

4. Look for the file ending in `.vmx` (e.g., `Windows 10 x64.vmx`).
5. Open it with the nano text editor by typing: `pluma "Windows 10 x64.vmx"` (use the quotes if there are spaces in the name).

```
(base) ┌──(bhai㉿bhai)-[~/vmware/Windows 10 x64]
└─$ sudo pluma 'Windows 10 x64.vmx'

(base) ┌──(bhai㉿bhai)-[~/vmware/Windows 10 x64]
└─$
```

6. Use your down arrow key to scroll to the absolute bottom of the file.
7. Paste these exact, clean lines:

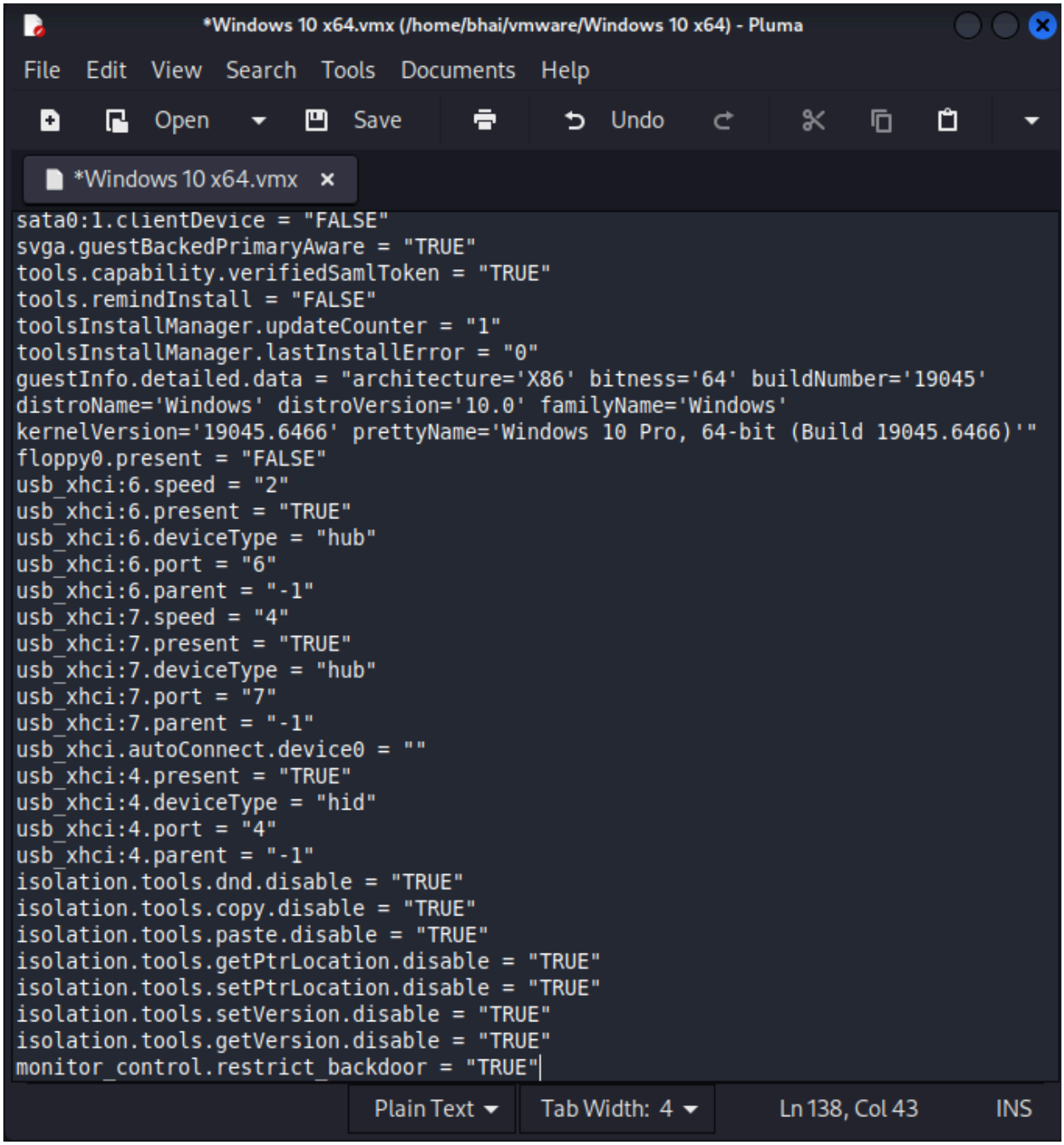
isolation.tools.getPtrLocation.disable = "TRUE"

isolation.tools.setPtrLocation.disable = "TRUE"

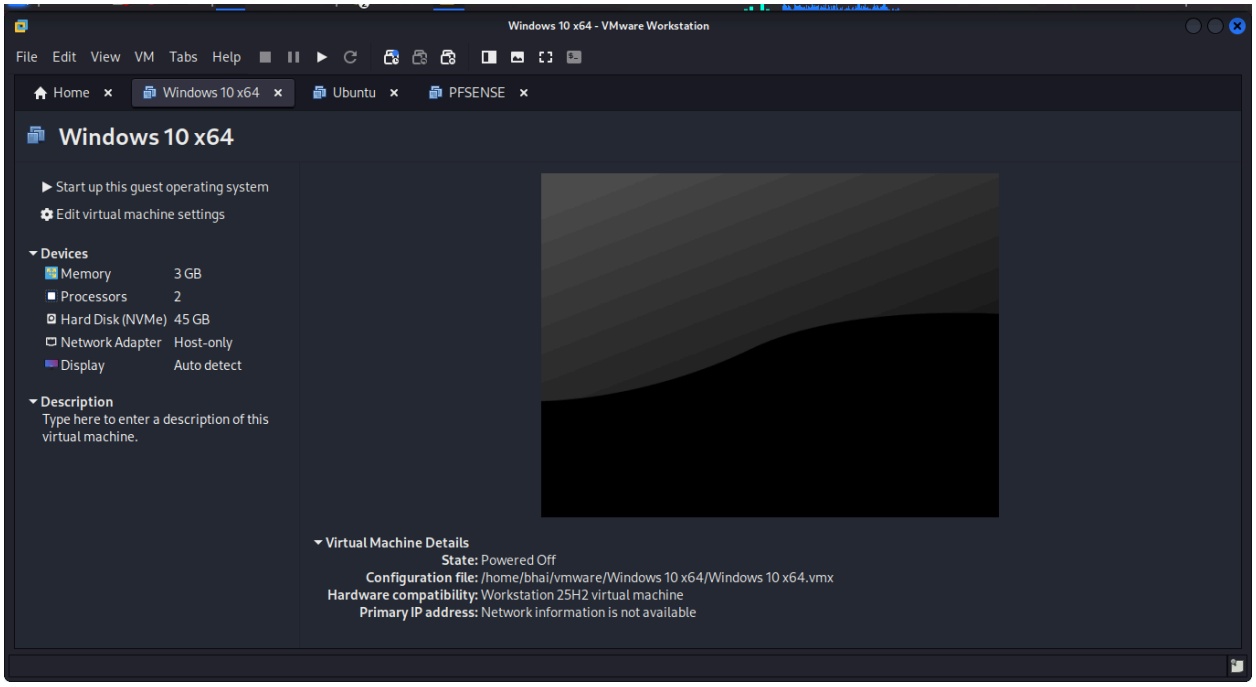
isolation.tools.setVersion.disable = "TRUE"

isolation.tools.getVersion.disable = "TRUE"

monitor_control.restrict_backdoor = "TRUE"



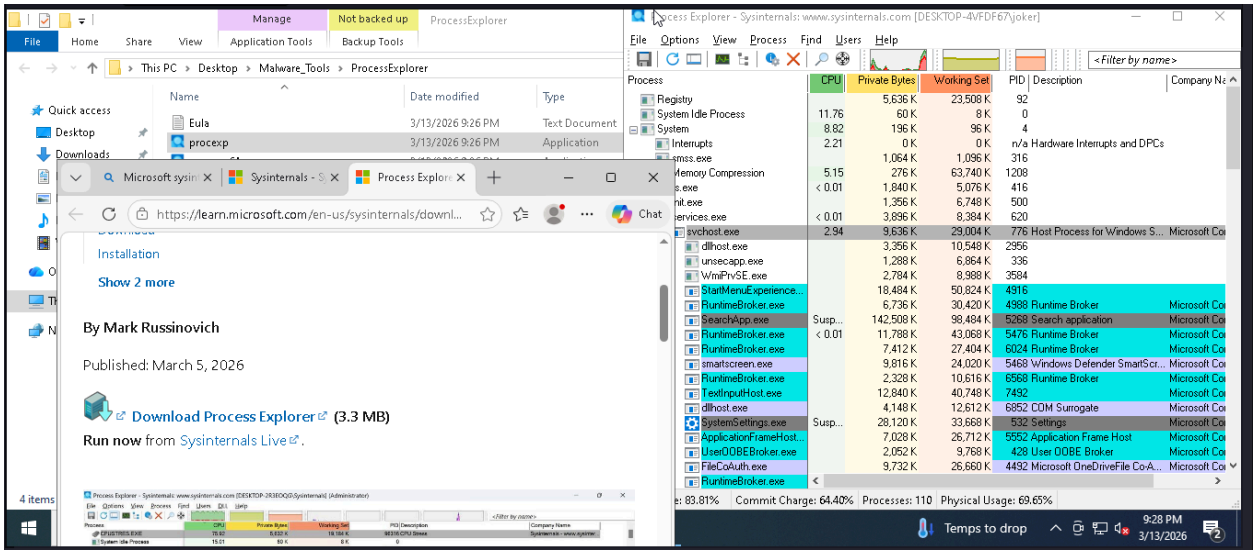
8. Press `Ctrl+O` then `Enter` to save, and `Ctrl+X` to exit nano. You are now cloaked!

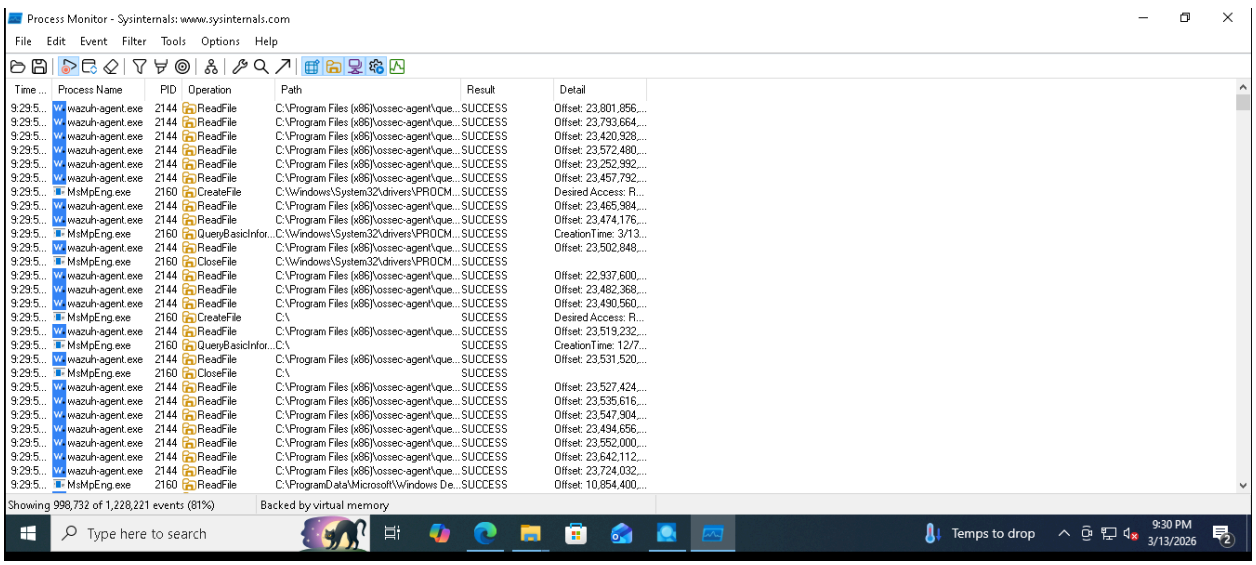


1. Process Explorer & Process Monitor (ProcMon)

These are official Microsoft Sysinternals tools. They do not need to be installed; they run as standalone executables.

- **Where to get them:** Go to `learn.microsoft.com/en-us/sysinternals/` (or just Google "Microsoft Sysinternals").
- **What to download:** Click on "Downloads" and grab both **Process Explorer** and **Process Monitor**. They will download as ZIP files. Extract both to a new folder on your desktop called `Malware_Tools`.

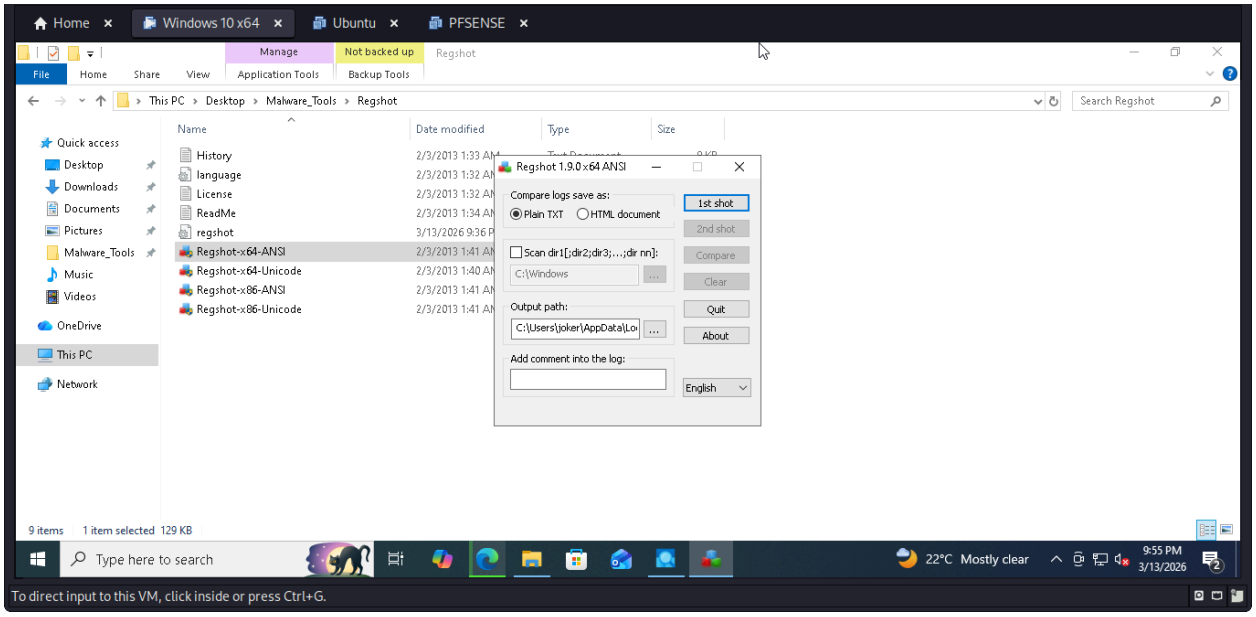




2. Regshot

This is an open-source tool that takes a snapshot of your registry before and after malware executes.

- **Where to get it:** Go to `sourceforge.net/projects/regshot/`.
- **What to download:** Click the big green download button for the latest version (usually a `.7z` or `.zip` file). Extract it into your `Malware_Tools` folder.

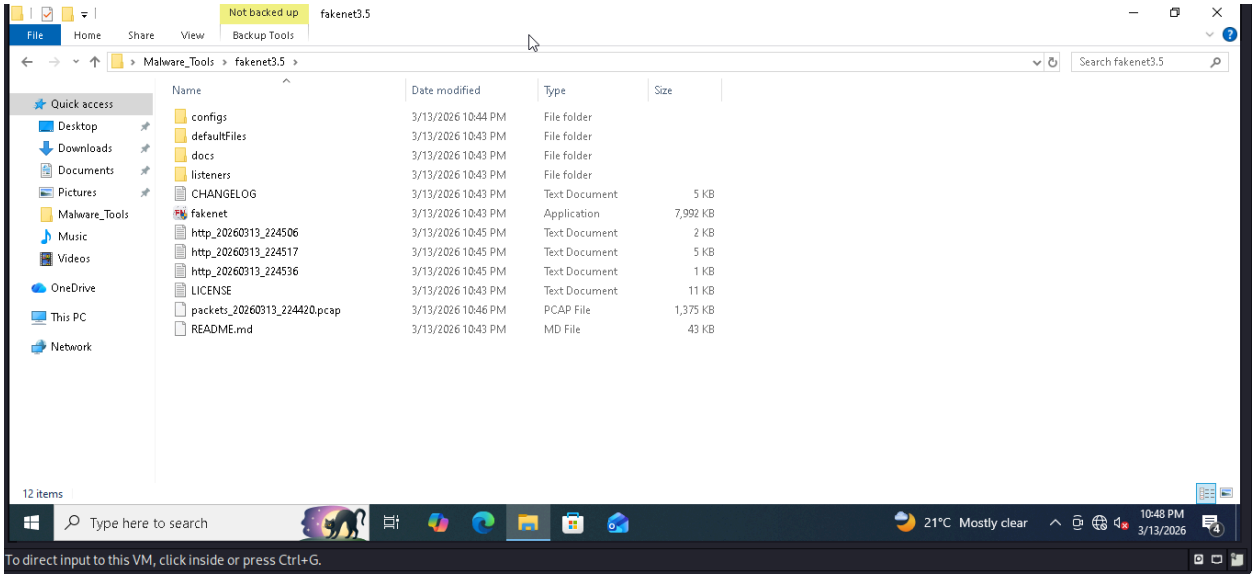


3. FakeNet-NG

This is the tool built by Mandiant/FireEye that will trick Agent Tesla into thinking it has an internet connection so we can steal its network traffic.

- **Where to get it:** Go to the official Mandiant GitHub repository: `github.com/mandiant/flare-fakenet-ng/releases`.
- **What to download:** Look for the compiled Windows release (it will be a `.zip` file under the "Assets" dropdown of the latest release). Extract it into your `Malware_Tools` folder.

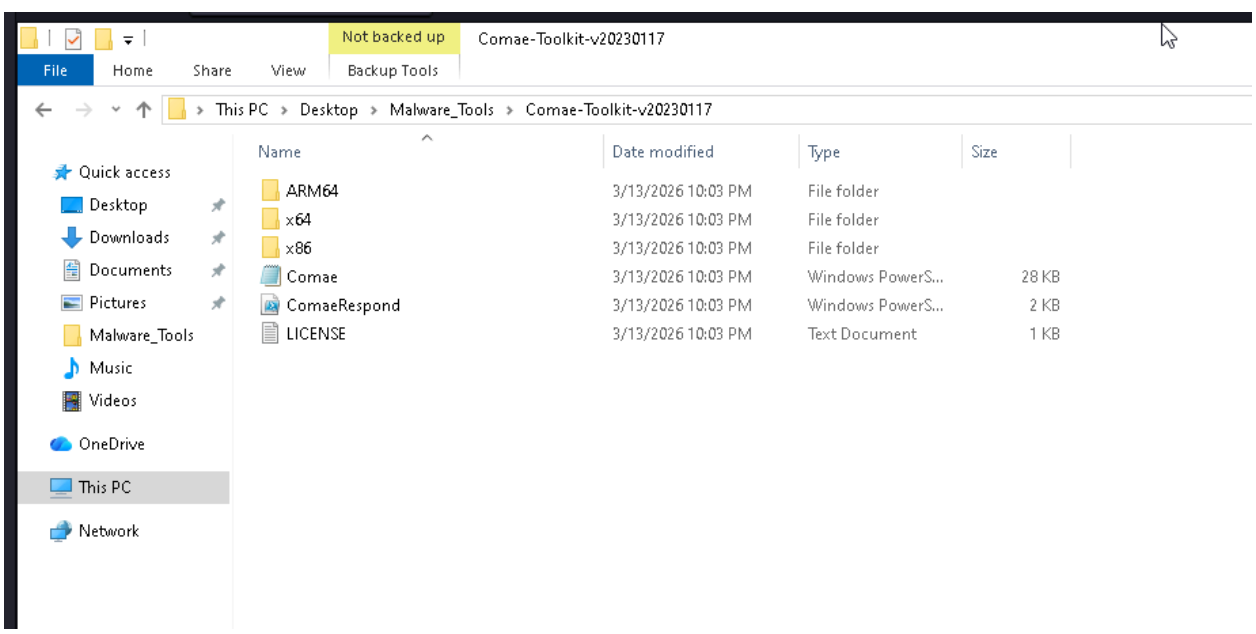
BUT IT WILL NOT SOMETIME INSTALL AND THE WINDOWS DEFENDER WILLL SCREAM IT CONTAINS VIRUS.. fOR THAT TURN OFF YOUR WINDOWS DEFENDER



4. DumpIt

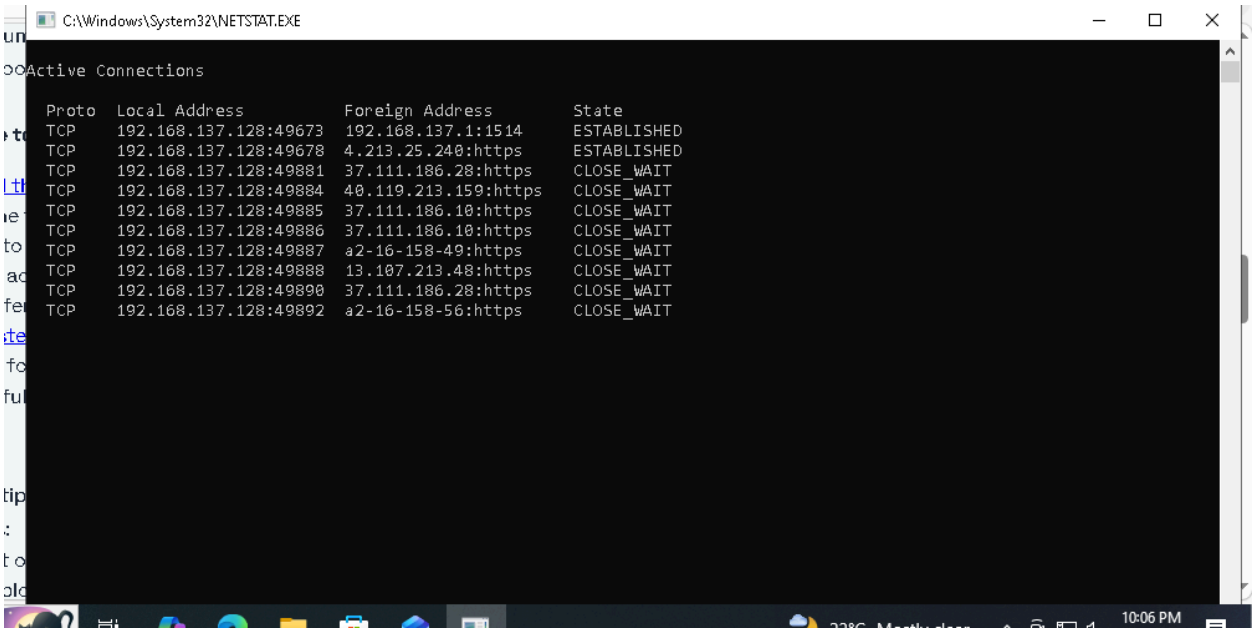
DumpIt is incredibly fast at capturing a computer's physical RAM for memory forensics.

- **Where to get it:** The original creator (Comae) was acquired by Magnet Forensics. You can get the official, updated "Magnet DumpIt for Windows" for free at `magnetforensics.com/free-tools/`.
- **What to download:** You will need to fill out a quick form with an email to get the download link. Alternatively, many analysts use the classic standalone `DumpIt.exe` mirrored on GitHub (like the `h4sh5/DumpIt-mirror` repository).



5. Netstat

Good news! You don't have to download anything for this. `netstat` is already built directly into Windows. We will just run it from the Command Prompt when the time comes.



How to Download Agent Tesla safely inside the VM

We are going to use **MalwareBazaar** (run by the Swiss security group abuse.ch). It is the absolute best place for analysts to get live malware. The best part? **They automatically package every download into a password-protected ZIP file for you!** (The password is always `infected`).

Here is the exact mission plan to get it right now before we cut the internet:

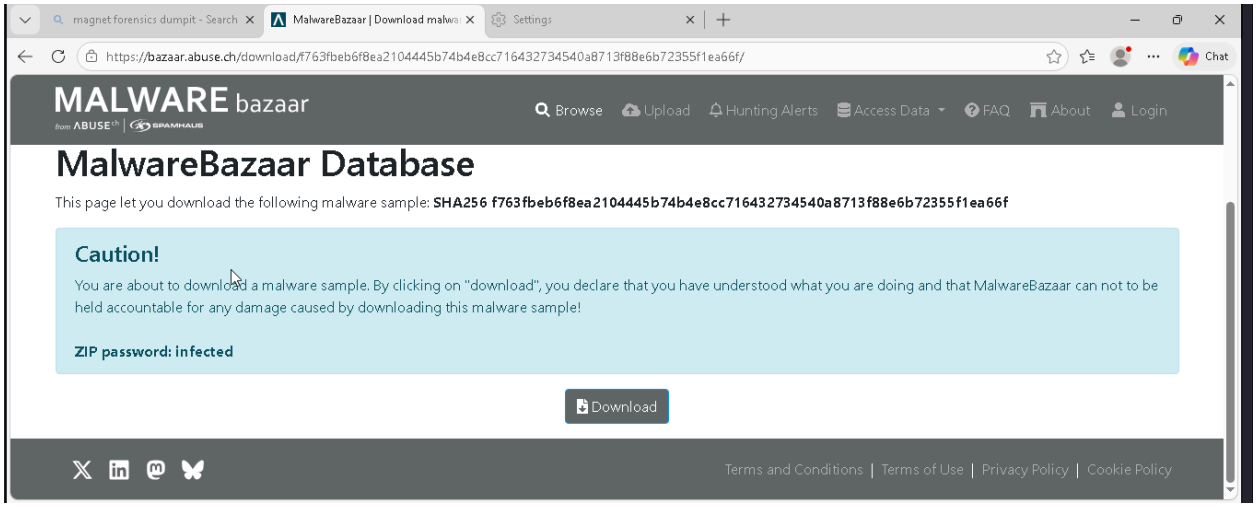
Step 1: Disable the Guards (Inside the VM)

1. Open the Windows Start Menu, type **Windows Security**, and open it.
2. Click **Virus & threat protection** -> **Manage settings**.
3. Turn **OFF** Real-time protection, Cloud-delivered protection, and Automatic sample submission. (You must do this or the download will fail immediately).

Step 2: Extract the Package

1. Open the Edge browser inside your VM.

2. Go to **`bazaar.abuse.ch`**.



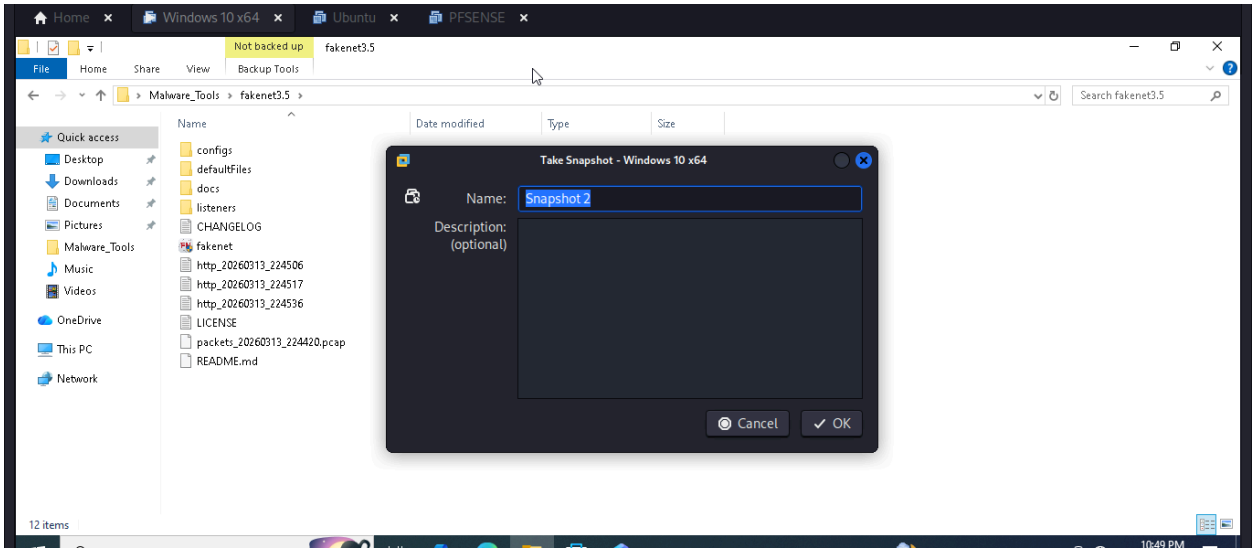
3. In the search bar at the top, type exactly this: `signature:AgentTesla` and hit enter.
4. You will see a list of thousands of Agent Tesla samples. Click on any recent one (preferably an `.exe` file, look at the tags).
5. On the sample's page, scroll down and click the **Download sample** button.
6. The file will download as a `.zip` archive. **DO NOT UNZIP IT YET.** Leave the ZIP file sitting safely on your Windows desktop.

Step 3: Burn the Bridge (Cut the Internet)

1. Now that the Agent Tesla ZIP file and all your tools are on the desktop, go back to your Kali machine's VMware settings.
2. Change the Network Adapter for the Windows 10 VM to **Host-Only**.

Step 4: The Final Snapshot

1. With the internet cut, the tools installed, and the ZIP file sitting safely on the desktop, go to your VMware menu.
2. Click **VM** -> **Snapshot** -> **Take Snapshot**.
3. Name it: `CLEAN BASELINE - READY TO DETONATE`.

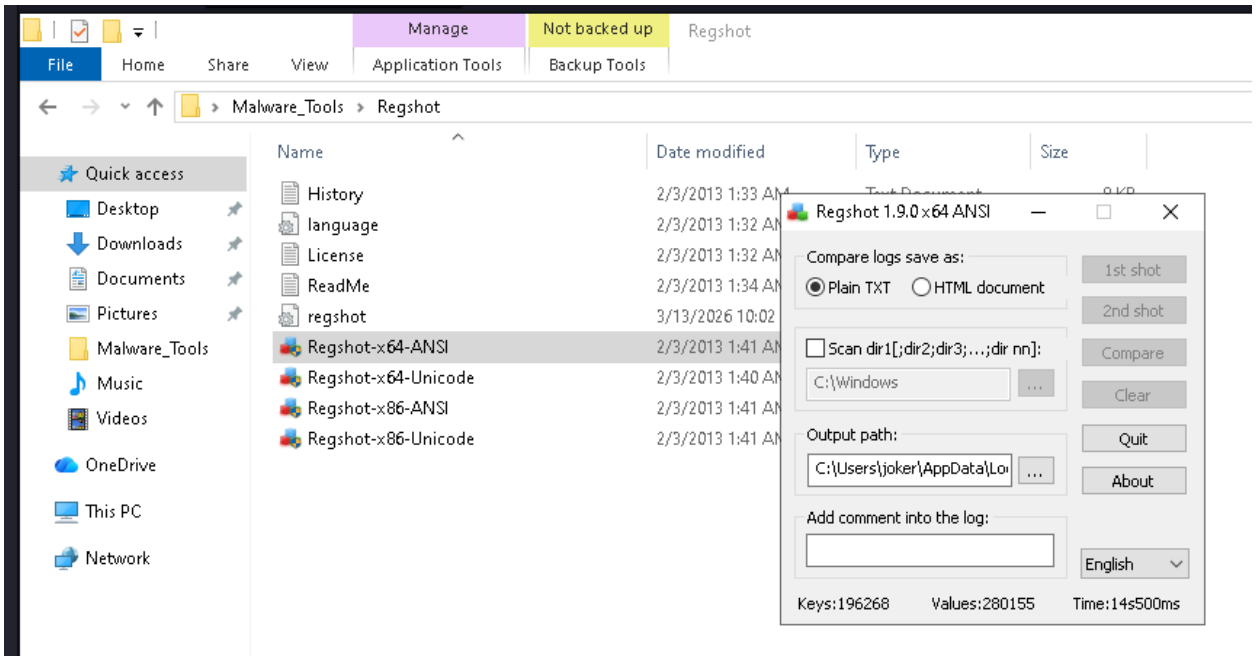


NOW Run every single tool as an Administrator.

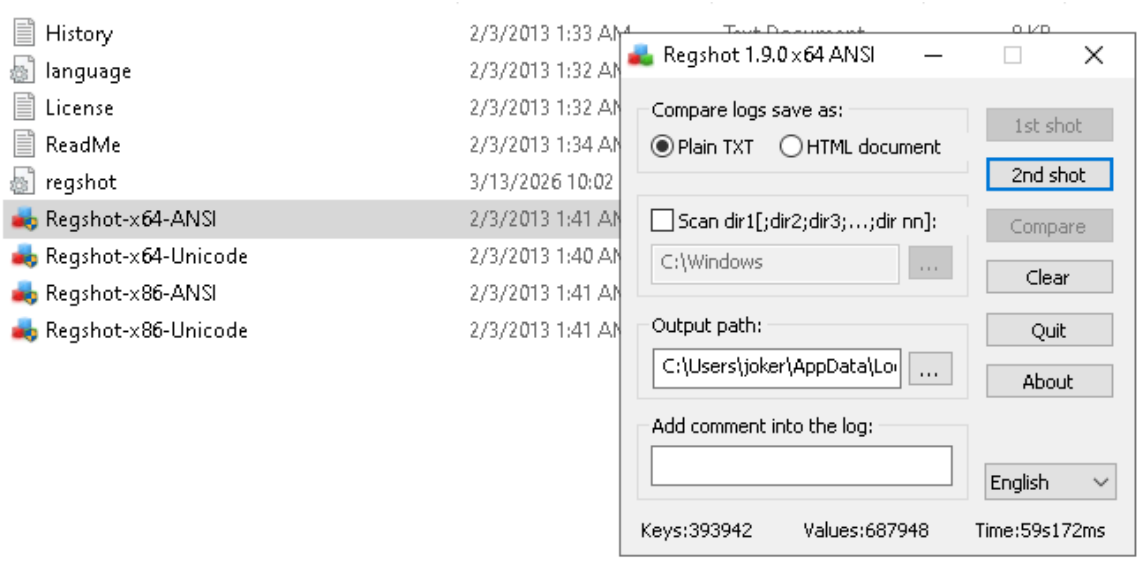
Phase 1: The Pre-Detonation Setup (Starting the Traps)

1. Baseline the Registry (Regshot)

1. Right-click `Regshot.exe` -> Run as Administrator.
2. Make sure the output format is set to Plain TXT.



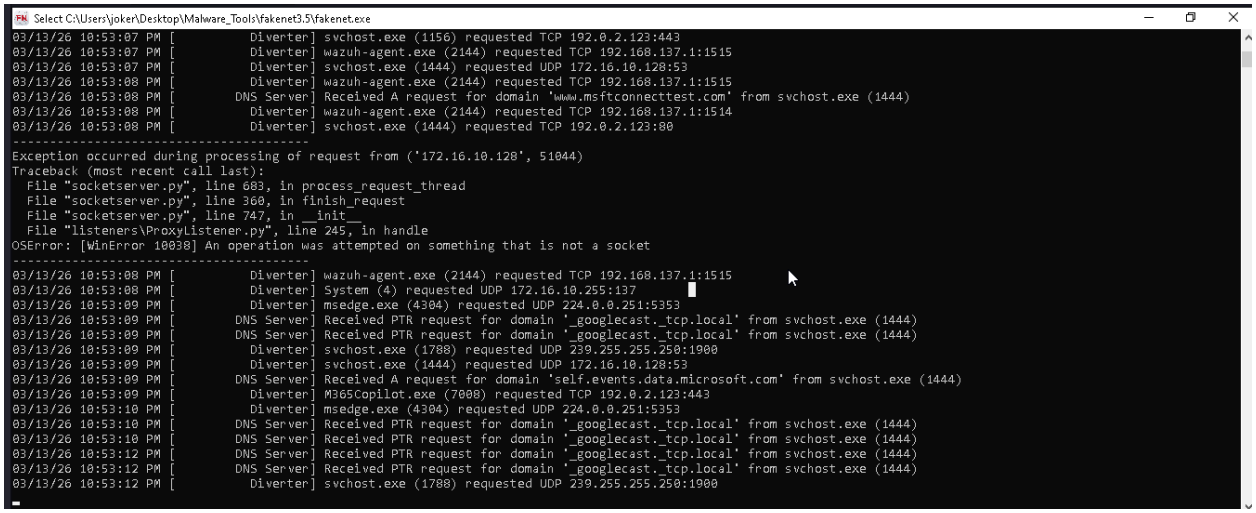
3. Click the 1st shot button, then click Shot.
4. The tool will freeze for a minute while it scans your entire registry. When it finishes, do not close the window. Just minimize it.



2. Trap the Network (FakeNet-NG & Netstat)

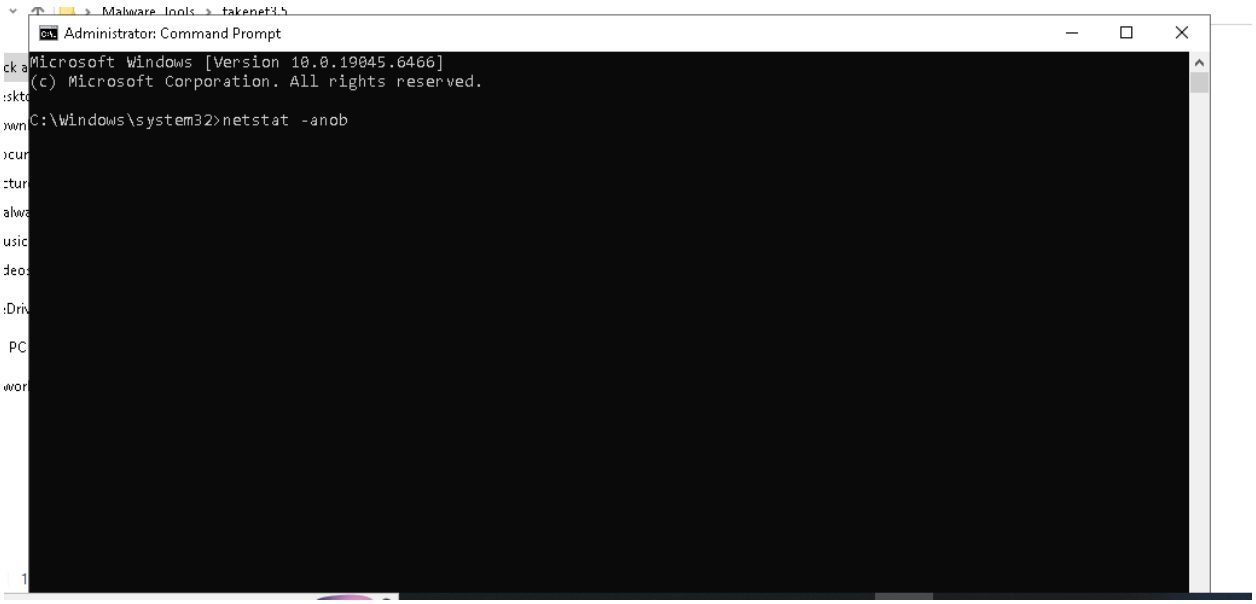
1. Right-click `fakenet.exe` -> Run as Administrator. A black console window will pop up and start initializing. It is now intercepting all fake internet traffic. Leave this window open

and visible.



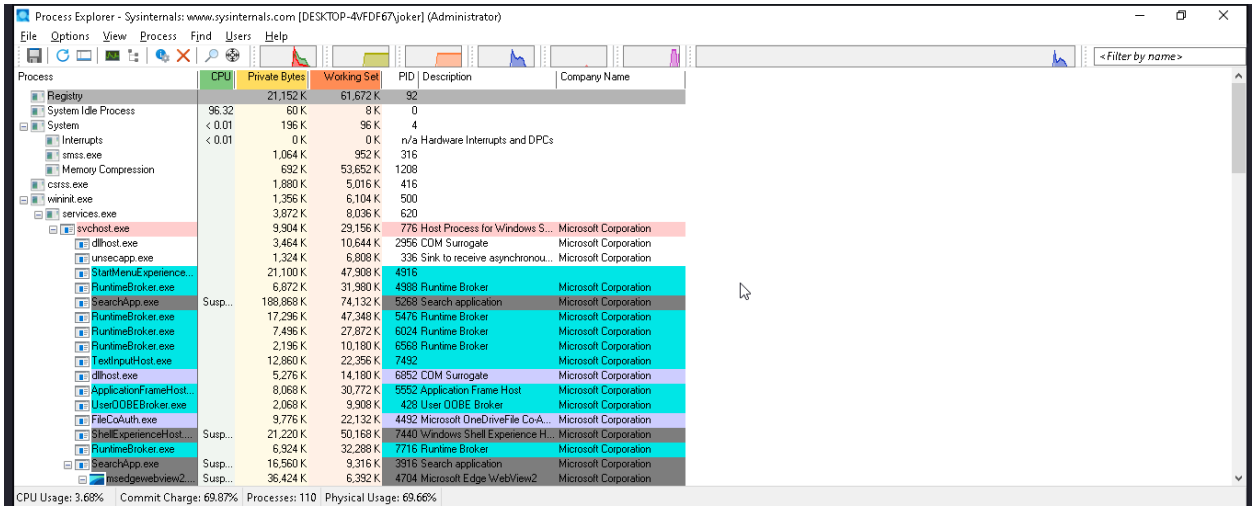
2. Open the Windows Start menu, type `cmd`, right-click Command Prompt, and select Run as Administrator.

3. Type `netstat -anob` but do not press Enter yet. Just leave the window ready.



3. Monitor the Brain (Process Explorer)

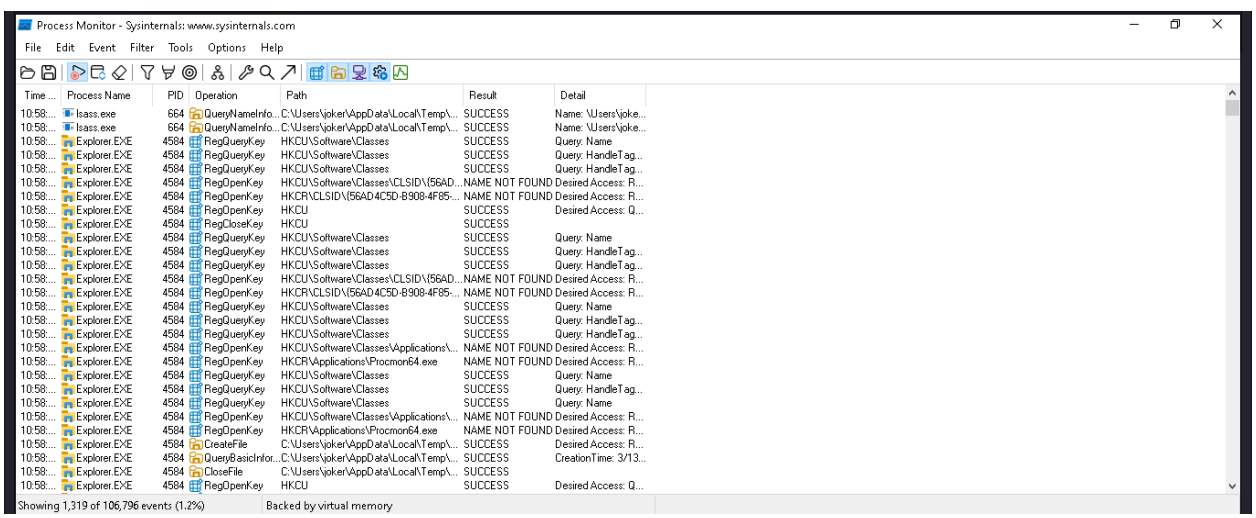
1. Right-click `procexp.exe` (Process Explorer) -> Run as Administrator.



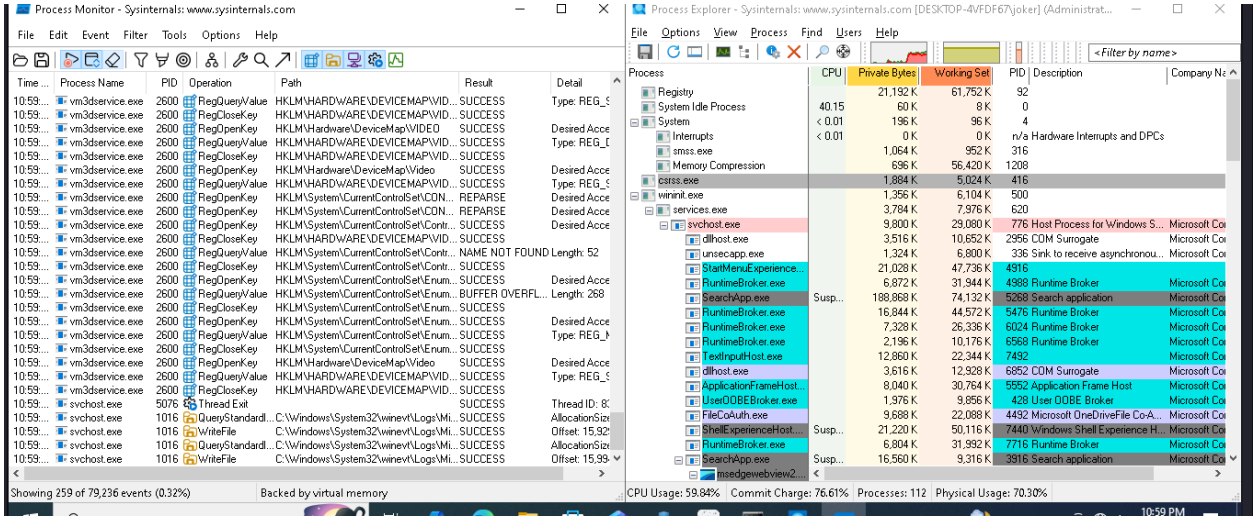
2. Click Options at the top and check Replace Task Manager.
3. Leave this window open on one side of your screen. This is where you will visually see Agent Tesla inject itself into other programs.

4. Record the File System (Process Monitor / ProcMon)

1. Right-click `Procmon.exe` -> **Run as Administrator**.
2. It will immediately start capturing millions of events (the screen will scroll like the Matrix).

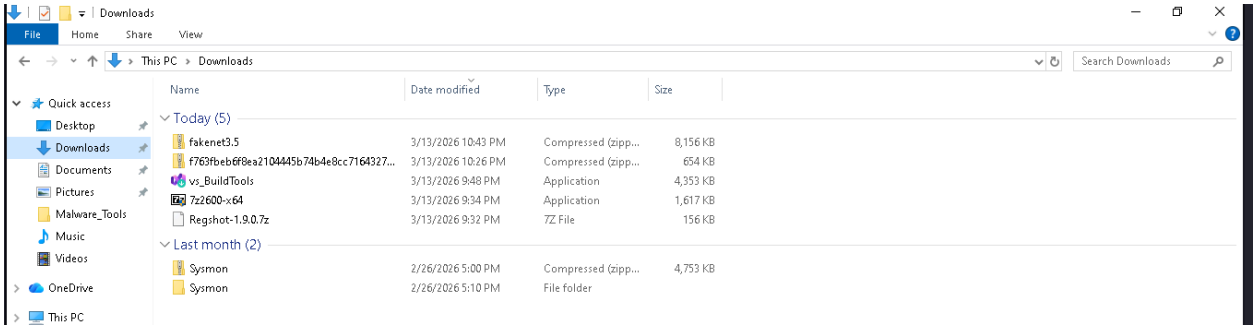


3. Right before we detonate the malware, click the **Clear** button (the little trash can icon at the top) or press `Ctrl+X`. This gives you a clean slate so you only record the malware's actions.



Phase 2: The Detonation

1. Right-click the Agent Tesla `.zip` file you downloaded from MalwareBazaar and extract it.

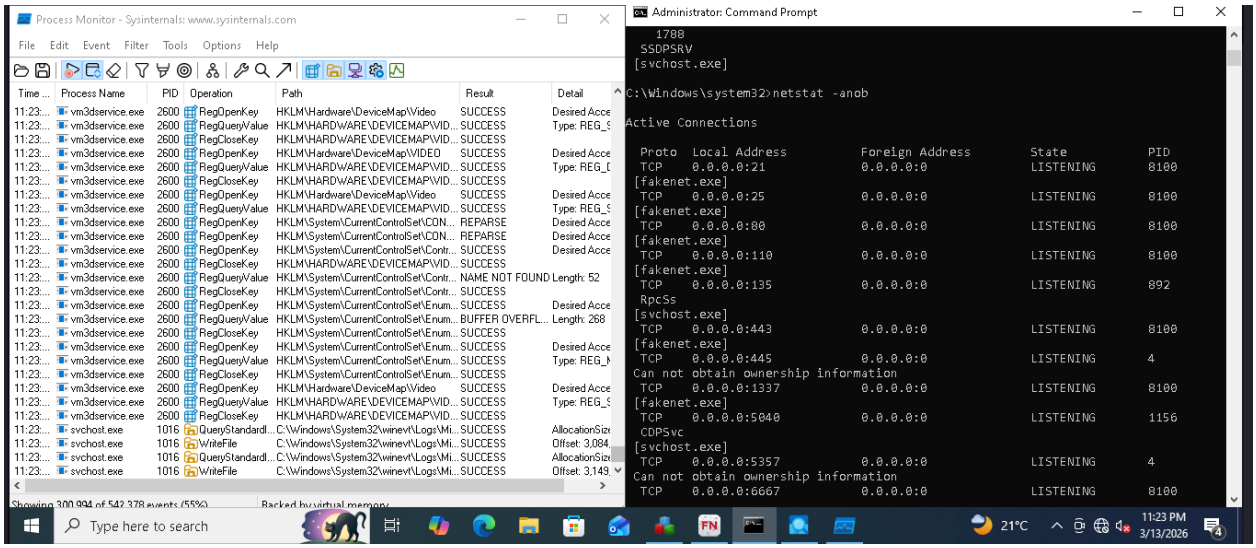


2. When it asks for a password, type: `infected`
3. You will see the raw malware executable. Take a deep breath.
4. **Double-click the malware executable.**

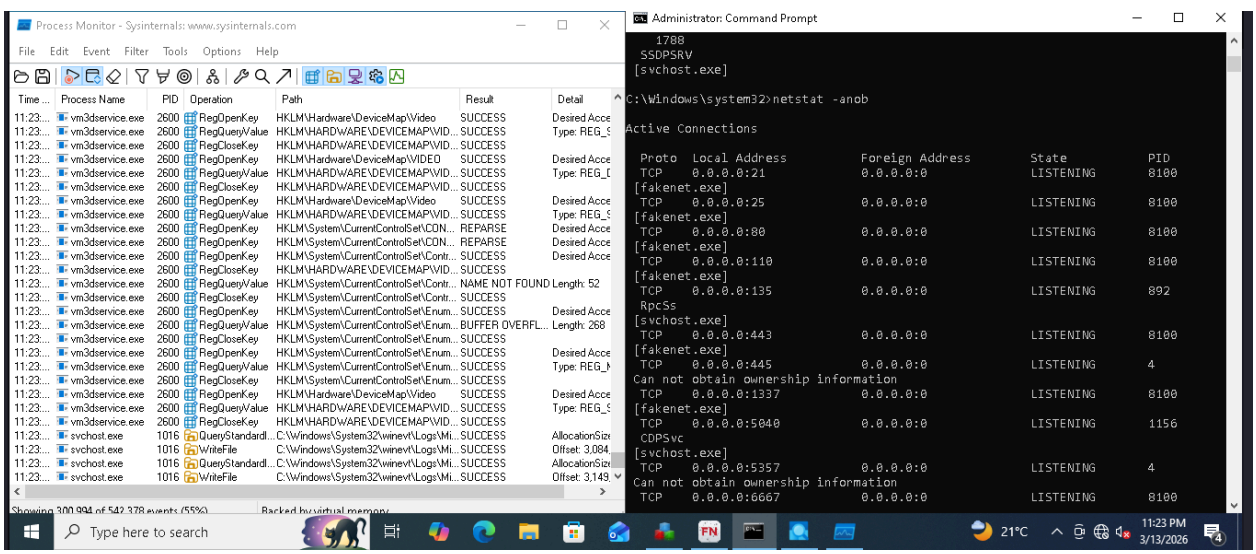
Phase 3: The Observation (Do not touch the mouse for 3 minutes!)

Advanced malware often "sleeps" for a minute to try and bypass antivirus sandboxes. Just watch your screens:

1. **Watch Process Explorer:** You should see the malware process pop up. Watch closely to see if it suddenly spawns a child process (like `cmd.exe`, `powershell.exe`, or `RegSvcs.exe`).



2. **Watch FakeNet-NG:** You will suddenly see lines of text appearing as Agent Tesla tries to do DNS lookups or send SMTP (email) traffic to exfiltrate data.
3. **Watch your Command Prompt:** Click on your CMD window and press **Enter** to run that `netstat -anob` command. Look for the malware's process ID establishing an outbound connection.



Phase 4: Data Collection (Securing the Evidence)

After about 3 to 5 minutes, the malware has done its damage. Now we collect the reports for your Task 4 PDF.

1. Stop Process Monitor (ProcMon)

1. Go to the ProcMon window. Click the magnifying glass icon at the top (or press `Ctrl+E`) to **Stop Capture**.
2. Click **File** -> **Save**. Save it as a CSV file to your desktop.

2. Take the Second Registry Shot (Regshot)

1. Bring up your minimized Regshot window.
2. Click the **2nd shot** button, then click **Shot**.
3. Wait for it to finish, then click the **Compare** button. Notepad will instantly pop open with a massive list of every single registry key the malware added or deleted. Save this Notepad file!

3. Dump the Memory (DumpIt)

1. Right-click `DumpIt.exe` -> **Run as Administrator**.
2. A console window will ask if you want to proceed. Type `y`.
3. It will take a few minutes to copy your entire RAM into a `.raw` file in the same folder.

4. Stop FakeNet-NG

1. Click on the FakeNet-NG black console window and press `Ctrl+C` to stop it.
2. It will automatically generate a `.pcap` network capture file in its folder.

Now, we have a classic malware analyst problem: **How do we get the data out without letting the malware slip through the door?**

Are the Evidence Files Infected?

First, let me reassure you: **Your evidence files are inherently safe.** Malware like Agent Tesla is an executable (`.exe`). It does not magically attach itself to plain text files (`.txt`), spreadsheet data (`.csv`), or network captures (`.pcap`). These are purely data files. As long as you do not

accidentally copy the actual Agent Tesla `.exe` file over to your Kali machine, your Kali machine remains 100% safe.

However, we practice strict lab hygiene. Here is the safest, most professional way to extract your evidence over your Host-Only network using built-in secure tools (no drag-and-drop required!).

Step 1: Package the Evidence (Inside the VM)

We want to move one single, neat package rather than dragging files around individually.

1. On your Windows 10 VM Desktop, create a new folder and name it `Task4_Evidence`.
2. Move your four evidence files (Regshot, ProcMon, DumpIt, and FakeNet) into this folder. **DO NOT put the malware `.exe` or its ZIP file in here.**
3. Right-click the `Task4_Evidence` folder -> **Send to** -> **Compressed (zipped) folder**. You now have a `Task4_Evidence.zip` file.

Step 2: Open the Secure Door (On Kali Linux)

We are going to use SSH (Secure Shell) to pull the file over the Host-Only network. Windows 10 actually has an SSH client built-in natively now!

1. Open a terminal on your Kali host.
2. Start your SSH service by running: `sudo systemctl start ssh`
3. Find your Kali machine's Host-Only IP address by typing: `ip a` (Look for the `vmnet` interface, usually something like `192.168.x.x`).

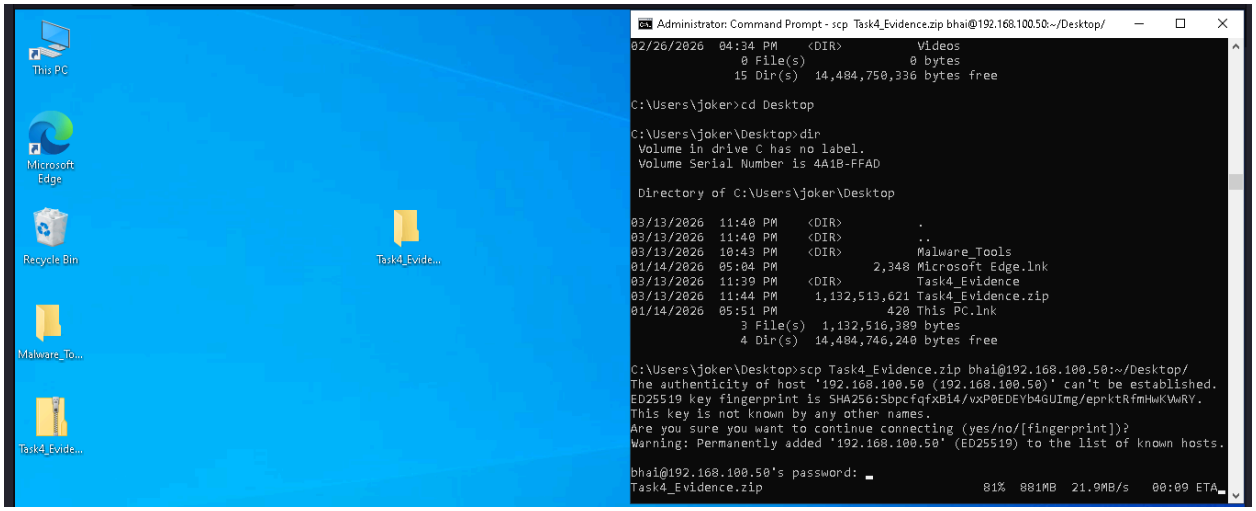
Step 3: Push the Evidence Out (Inside the VM)

Now we are going to securely push the ZIP file from the infected VM straight onto your Kali desktop.

1. Inside your Windows 10 VM, open the **Command Prompt**.
2. Navigate to your desktop by typing: `cd Desktop`

Use the native `scp` (Secure Copy) command to push the file to Kali. Type this exactly, replacing `kali` with your actual Kali username, and the IP with your Kali IP:
DOS

scp Task4_Evidence.zip kali@192.168.x.x:~/Desktop/

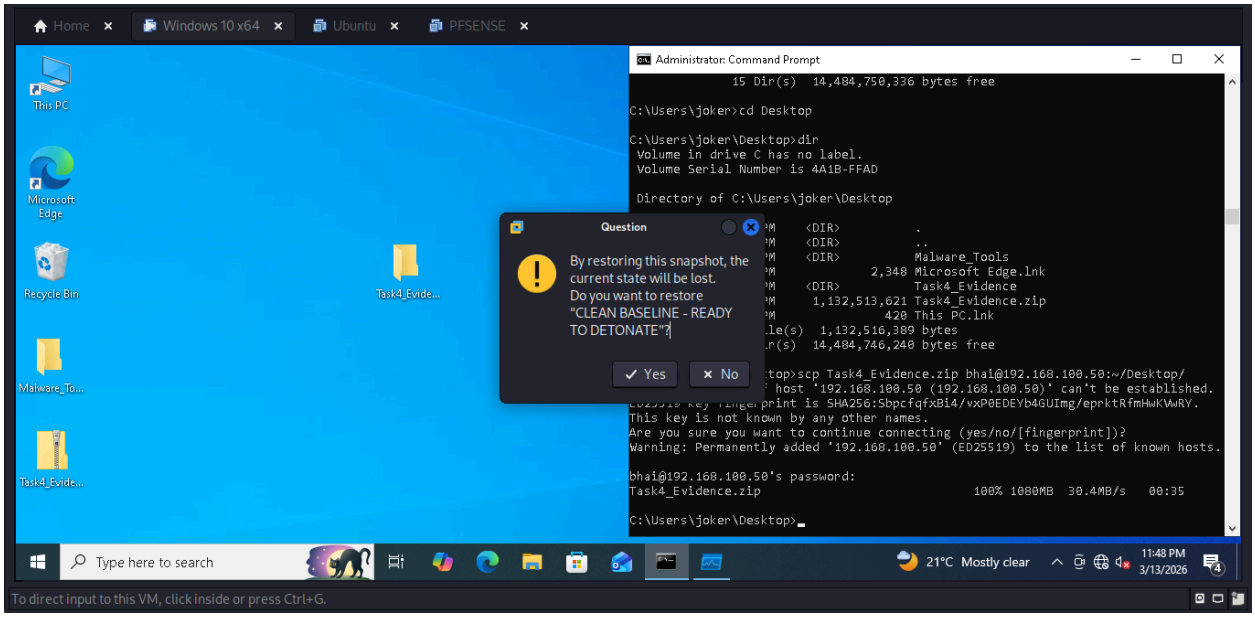
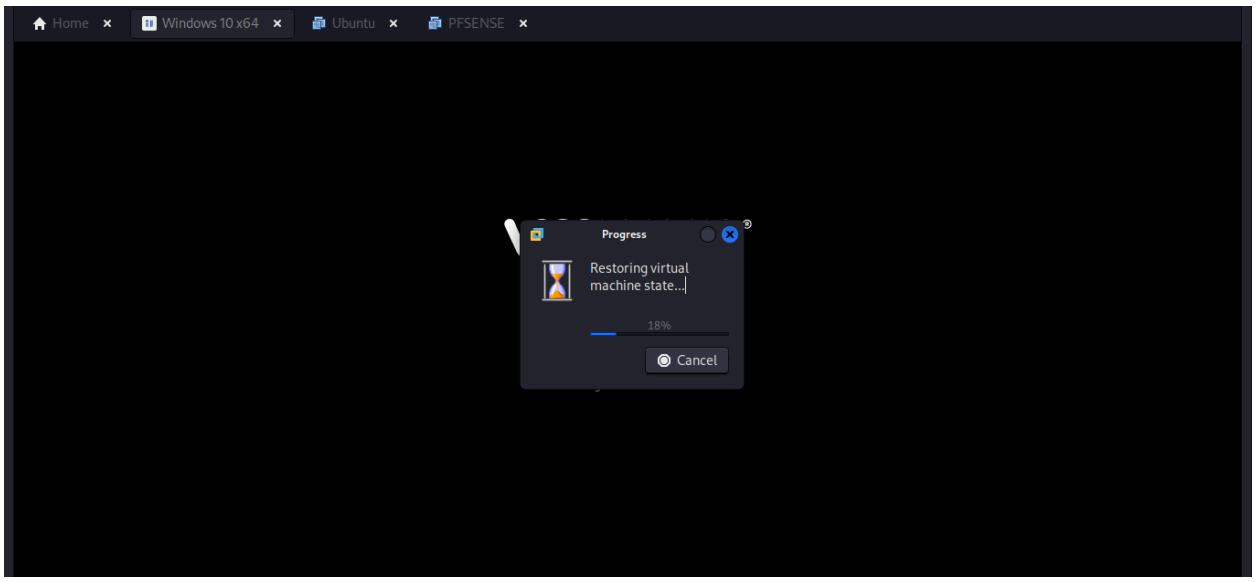
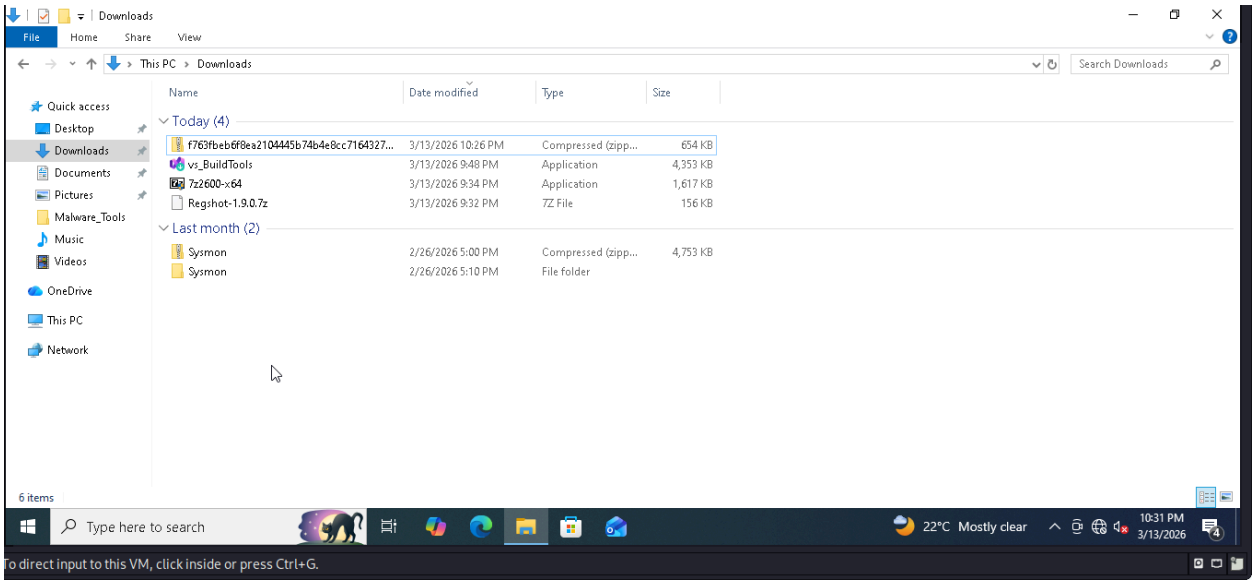


3. It will ask if you trust the fingerprint (type `yes`), and then it will ask for your Kali password. Type it in and hit Enter.
4. You will see a transfer progress bar. Once it hits 100%, the file is safely on your Kali desktop!

Step 4: NUKE IT FROM ORBIT (The Revert)

The second you verify that `Task4_Evidence.zip` is sitting safely on your Kali desktop, it is time to destroy the infected environment.

1. Go to your VMware top menu bar.
2. Click **VM** -> **Snapshot** -> **Revert to Snapshot** (Select the `CLEAN BASELINE` snapshot you made earlier).



3. VMware will instantly wipe the RAM and hard drive state, erasing Agent Tesla from existence and returning your VM to a perfect, sparkling clean state in about 5 seconds.